

Asaya Weston
Advanced Programming
Reflection Report
GitHub Link: https://github.com/ccybersai/airbnb_explorer

For this project, I built a Python command-line application that explores Airbnb listings data from Bangkok, Thailand. From Inside Airbnb, I got the listings csv for Bangkok. There are 31,069 listings with the prices being in Thai baht. I implemented three methods. `get_summary` gives an overview of the dataset, `search_entries` finds listings by neighborhood or price range, and `get_aggregate` calculates average and median prices by neighborhood or room type.

The first thing I did was set up the project structure, virtual environment, `requirements.txt`, and `.gitignore`. I then built `data_handler.py`, which covered cleaning and the core logic, and tested it on the real data. The data needed a lot of cleaning. Prices were stored as text like "\$1,148.24," 2,082 listings had no price, and some were extreme outliers, like one at 25,106 THB a night. I converted prices to numbers, dropped incomplete rows, and removed outliers above 7,034 THB using Tukey's rule, which left 27,566 listings. I added `visualization.py`, then `cli.py` and `main.py`. I wrote 13 unit tests with `pytest`, all passing. I committed after each working piece and pushed to GitHub. I also ran into Git setup problems. I had to install Apple's Command Line Tools to get Git, replace placeholder name and email info on my first commits, and create a personal access token because GitHub doesn't accept regular passwords for pushing.

A big problem I ran into was my program hanging on startup. When I pressed Ctrl + C, the traceback showed that Python got stuck reading files while importing Plotly. Working through it with Claude, I found that the cause was iCloud syncing the Documents folder and offloading venv files. I rebuilt the environment as `venv.nosync`, so that iCloud would ignore it, and then linked it back as `venv`, and updated `.gitignore`.

Some insights I found from my data are that Pathum Wan is the most expensive district with an average of 2,692 THB. Vadhana, Khlong Toei, and Huai Khwang have the most listings. Entire homes make up about two-thirds of listings. Shared rooms are much cheaper. About 45% of listings cost 500–1,500 THB a night. A lot of columns, like `neighbourhood_group_cleansed`, `calendar_updated`, `host_since`, and `license`, are missing in all 27,566 rows, which means Inside Airbnb doesn't collect them for Bangkok. Review scores are missing for about a third of the listings, likely the ones with no reviews yet.

This project reminded me that real-world data is rarely ready to use. Cleaning the prices and removing outliers had a big effect on the results, as I expected. Building and testing each piece separately made bugs easier to find, since I always knew which piece had just changed. If I did it again, I would keep my project outside iCloud-synced folders from the start, which would have saved me the time spent figuring out why Python kept freezing.

CLI OUTPUTS

python main.py --search neighborhood "khlong toei"

```
=====
SEARCH: neighborhood = 'khlong toei'
=====
Found 3,492 matching listings. Showing first 20:
```

ID	Name	Neighborhood	Room type	Price
979584324955463360	厨房可做饭免费Wi-Fi健身房泳池800米到坚果地铁站旅游打卡地	Khlong Toei	Entire home/apt	1,148
1519261399728812558	Em District staycation 420 貴婦商場	Khlong Toei	Entire home/apt	2,061
41439202	Apartment Sukhumvit 30 (500m. to Em	Khlong Toei	Entire home/apt	1,370
8765558	JW Serviced Apartment for Rent #7	Khlong Toei	Private room	1,390
1105816668065171912	Sukhumvit 50 One bedroom quiet nice	Khlong Toei	Entire home/apt	1,157
42978873	tara Rama4/ true coffee	Khlong Toei	Entire home/apt	4,143
1545505324101062254	Chill Inn Bangkok-Double Room	Khlong Toei	Private room	1,299
1338177587983604112	Modern Luxury 2 Bedrooms Apt The Lu	Khlong Toei	Entire home/apt	1,734
566364418184313097	5min BTS,NiceDeluxeRoom4@Sukhumvit&	Khlong Toei	Private room	1,085
1683198275767209209	Center/Private Bedroom&Bathroom/4th	Khlong Toei	Private room	1,573
1438086649148961059	Condo close to BTS Phraknong	Khlong Toei	Private room	653
53960529	【Bright & Quiet】near Subway, Smart	Khlong Toei	Entire home/apt	1,980
1217122750479400651	One-Bedroom Suite By Neom Town Resi	Khlong Toei	Private room	2,117
1560018081898450746	Deluxe Studio @Omni Complex, Nana	Khlong Toei	Private room	1,302
23835175	Rooftop pool, gym, luxury cozy 1BR	Khlong Toei	Entire home/apt	1,674
735236097943535169	New! Gateway/ Bangkok University 1B	Khlong Toei	Entire home/apt	1,253
902675895377070408	Deluxe Room with Balcony	Khlong Toei	Private room	2,220
1215531043136178780	3Beds 6Pax Entire Apt 2Bath BTSAsok	Khlong Toei	Entire home/apt	5,082
643954796060369240	Brand New Studio Near MRT Sirikit (	Khlong Toei	Private room	805
1305711807384537529	Nana Family Room	Khlong Toei	Entire home/apt	4,627

python main.py --aggregate neighborhood

```
(venv)root@root:~/asayaweston/asayas-facebook-ai-df# python main.py --aggregate neighborhood
2026-10-02 20:10:39,912 - INFO - Loaded dataset with 31069 rows
2026-10-02 20:10:39,957 - INFO - Cleaning removed 3503 rows; 27566 remain (price cap 7,034)
```

PRICE STATISTICS BY NEIGHBORHOOD (THB)			
Neighborhood	Mean	Median	Listings
Parthum Wan	2,692	2,339	614
Vadhana	2,449	2,071	3,766
Samphanthawong	2,425	2,054	291
Bang Rak	2,298	1,929	975
Pom Prap Sattru Phai	2,149	1,710	252
Buang Kum	2,096	2,031	241
Khlong Toei	2,079	1,778	3,492
Bangkok Yai	2,060	1,586	138
Khlong Sam Wa	2,056	1,028	29
Ratchathewi	2,038	1,722	1,642
Thawi Watthana	2,010	1,308	20
Phaya Thai	1,991	1,541	470
Lat Phrao	1,966	2,064	73
Huai Khwang	1,871	1,672	3,653
Bang Khun thain	1,824	948	20
Pra Wet	1,822	1,170	188
Nong Khaem	1,820	660	8
Din Daeng	1,810	1,367	550
Sathon	1,806	1,369	1,187
Sai Mai	1,765	980	64
Taling Chan	1,742	1,401	51
Lat Krabang	1,735	1,286	254
Khlong San	1,715	1,274	685
Bang Kho laen	1,699	1,285	205
Phra Khanong	1,687	1,408	1,032
Phra Nakhon	1,632	1,310	1,297
Dusit	1,625	1,320	113
Thung khru	1,601	1,119	26
Khan Na Yao	1,591	1,092	71
Wang Thong Lang	1,579	1,022	150
Chatu Chak	1,567	1,198	784
Bang Khen	1,551	970	160
Bangkok Noi	1,541	1,210	207
Chom Thong	1,504	1,063	60
Saphan Sung	1,494	868	91
Rat Burana	1,489	1,170	54
Min Buri	1,476	1,200	77
Yan na wa	1,471	1,185	402
Don Mueang	1,445	1,034	232
Nong Chok	1,420	838	17
Bang Kapi	1,397	1,095	565
Bang Khae	1,373	1,096	57
Phasi Charoen	1,368	1,165	190
Bang Bon	1,327	908	12
Suan Luang	1,321	1,057	794
Lak Si	1,265	999	149
Bang Sue	1,236	1,000	409
Bang Na	1,235	985	793
Thon buri	1,169	949	510
Bang Phlat	1,093	741	446

python main.py --export budget.csv --search price 500-1500 --columns id name neighborhood room_type price, which saved 12,447 listings priced 500–1

```
(venv.nosync) asayaweston@Asayas-MacBook-Air airbnb_explorer % python main.py --export budget.csv --search price 500-1500 --columns id name neighborhood room_type price
2026-10-02 20:11:32,763 - INFO - Loaded dataset with 31069 rows
2026-10-02 20:11:32,836 - INFO - Cleaning removed 3503 rows; 27566 remain (price cap 7,034)
2026-10-02 20:11:32,859 - INFO - Exported 12447 rows to data/processed/budget.csv

=====
EXPORT COMPLETE
=====
Filter: price = '500-1500'
Columns: id, name, neighborhood, room_type, price
Saved 12,447 rows to data/processed/budget.csv
```

python main.py --aggregate room_type

```
(venv.nosync) asayaweston@Asayas-MacBook-Air airbnb_explorer % python main.py --aggregate room_type
2026-10-02 20:10:17,809 - INFO - Loaded dataset with 31069 rows
2026-10-02 20:10:17,849 - INFO - Cleaning removed 3503 rows; 27566 remain (price cap 7,034)

=====
PRICE STATISTICS BY ROOM_TYPE (THB)
=====
```

Room Type	Mean	Median	Listings
Hotel room	2,039	1,927	256
Entire home/apt	2,013	1,643	18,178
Private room	1,691	1,369	8,636
Shared room	599	430	496

python -m pytest -v. All 13 unit tests for DataHandler and Visualization passed.

```
(venv.nosync) asayaweston@Asayas-MacBook-Air airbnb_explorer % python -m pytest -v
===== test session starts =====
platform darwin -- Python 3.14.7, pytest-9.1.1, pluggy-1.6.0 -- /Users/asayaweston/Documents/classes/cis301/airbnb_explorer/venv.nosync/bin/python
cachedir: .pytest_cache
rootdir: /Users/asayaweston/Documents/classes/cis301/airbnb_explorer
plugins: platformdirs-4.12.2
collected 13 items

tests/test_data_handler.py::test_price_converted_to_numeric PASSED [ 7%]
tests/test_data_handler.py::test_rows_with_missing_values_dropped PASSED [ 15%]
tests/test_data_handler.py::test_neighborhood_column_renamed PASSED [ 23%]
tests/test_data_handler.py::test_search_is_case_insensitive PASSED [ 30%]
tests/test_data_handler.py::test_search_price_range PASSED [ 38%]
tests/test_data_handler.py::test_aggregate_mean_price PASSED [ 46%]
tests/test_data_handler.py::test_export_selected_columns PASSED [ 53%]
tests/test_data_handler.py::test_invalid_column_raises PASSED [ 61%]
tests/test_data_handler.py::test_missing_file_raises PASSED [ 69%]
tests/test_visualization.py::test_output_dir_created PASSED [ 76%]
tests/test_visualization.py::test_missing_price_column_returns_none PASSED [ 84%]
tests/test_visualization.py::test_price_distribution_saves_png PASSED [ 92%]
tests/test_visualization.py::test_neighborhood_plot_saves_jpg PASSED [100%]

===== 13 passed in 4.20s =====
```

```
python main.py --summary
```

```

levy@hoyinc: /$ jupyter lab --ip 0.0.0.0 --port 8888 --no-browser --allow-root
2026-10-02 20:04:26,625 - INFO - Loaded dataset with 31069 rows
2026-10-02 20:04:26,664 - INFO - Cleaning removed 3583 rows; 27566 remain (price cap 7,034)

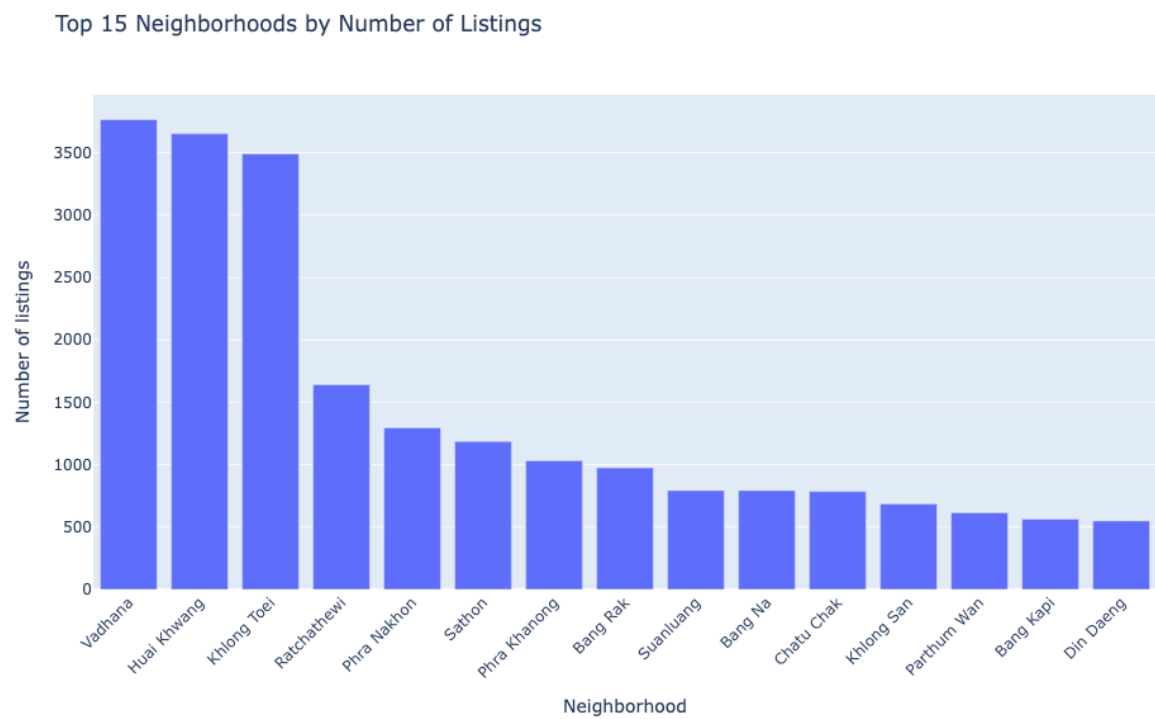
=====
DATASET SUMMARY
=====
Rows before cleaning: 31,069
Rows after cleaning: 27,566
Columns (89):
    id, listing_url, scrape_id, last_scraped, source, name, description, neighborhood_overview, picture_url, host_id, host_url, host_profile_id, host_profile_url, host_name, host_sin
    ce_time, time_as_user, years, hosts_time_as_user, years, hosts_time_as_host_months, host_location, host_about, host_response_time, host_response_rate, hos
    t_acceptance_rate, host_is_superhost, host_thumbnail_url, host_picture_url, host_neighbourhood, host_listings_count, host_total_listings_count, host_verifications, host_has_profile
    pic, host_identity_verified, neighborhood, neighbourhood_group_cleaned, latitude, longitude, property_type, room_type, accommodates, bathrooms, bathrooms_text, bedrooms, beds, am
    enities, price, price_quote_checkin_date, price_quote_checkout_date, price_quote_total_price, price_quote_price_per_night, price_quote_raw, minimum_nights, maximum_nights, minimum
    _nights, maximum_minimum_nights, minimum_maximum_nights, maximum_maximum_nights, minimum_nights_avg_ntm, maximum_nights_avg_ntm, calendar_updated, has_availability, availabi
    lity_30, availability_60, availability_90, availability_365, calendar_last_scraped, number_of_reviews, number_of_reviews_ltm, number_of_reviews_l30d, availability_eoy, number_of_re
    views, estimated_occupancy_1365d, estimated_revenue_1365d, first_review, last_review, review_scores_rating, review_scores_accuracy, review_scores_cleanliness, review_scores_chec
    kin, review_scores_communication, review_scores_location, review_scores_value, license, instant_bookable, calculated_host_listings_count, calculated_host_listings_count_entire_home
    s, calculated_host_listings_count_private_rooms, calculated_host_listings_count_shared_rooms, reviews_per_month

Columns with missing values (37):
neighborhood_overview      27,566
host_since                 27,566
host_response_time        27,566
host_response_rate        27,566
host_acceptance_rate      27,566
host_thumbnail_url        27,566
host_neighbourhood        27,566
host_total_listings_count 27,566
host_verifications        27,566
neighbourhood_group_cleaned 27,566
calendar_updated          27,566
license                   27,566
instant_bookable          27,566
host_about                10,636
review_scores_value       8,859
review_scores_checkin     8,858
review_scores_location    8,858
review_scores_communication 8,857
first_review              8,856
last_review               8,856
review_scores_rating      8,856
review_scores_accuracy    8,856
review_scores_cleanliness 8,856
reviews_per_month         8,856
host_location             7,117
bedrooms                  3,505
bathrooms                 2,619
beds                      1,222
description               384
bathrooms_text            123
has_availability          6
minimum_nights            2
maximum_nights            2
minimum_minimum_nights    2
maximum_minimum_nights    2
minimum_maximum_nights    2
maximum_maximum_nights    2

```

VISUALIZATIONS

Bar chart of the top 15 neighborhoods by number of listings



Price distribution histogram

